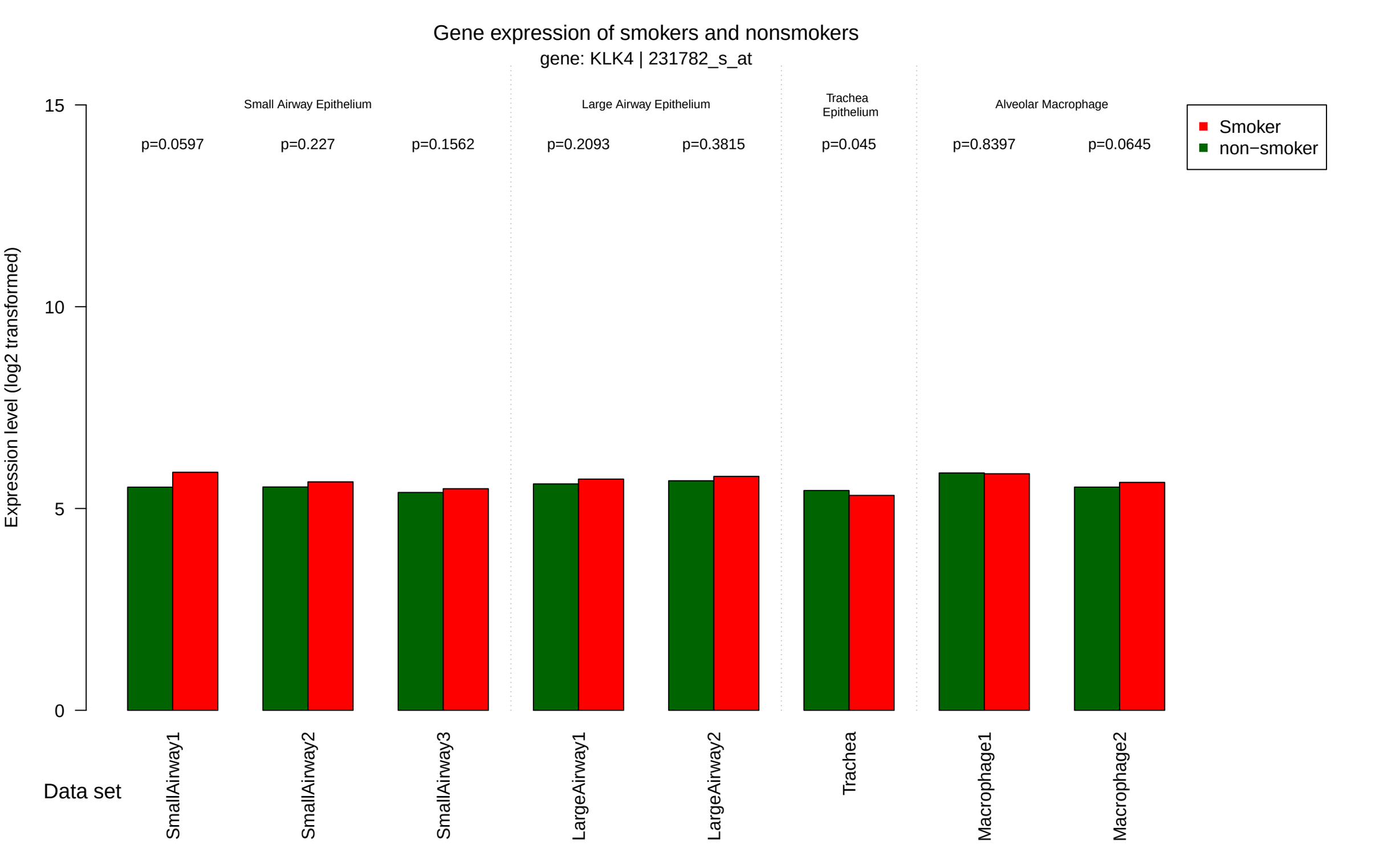
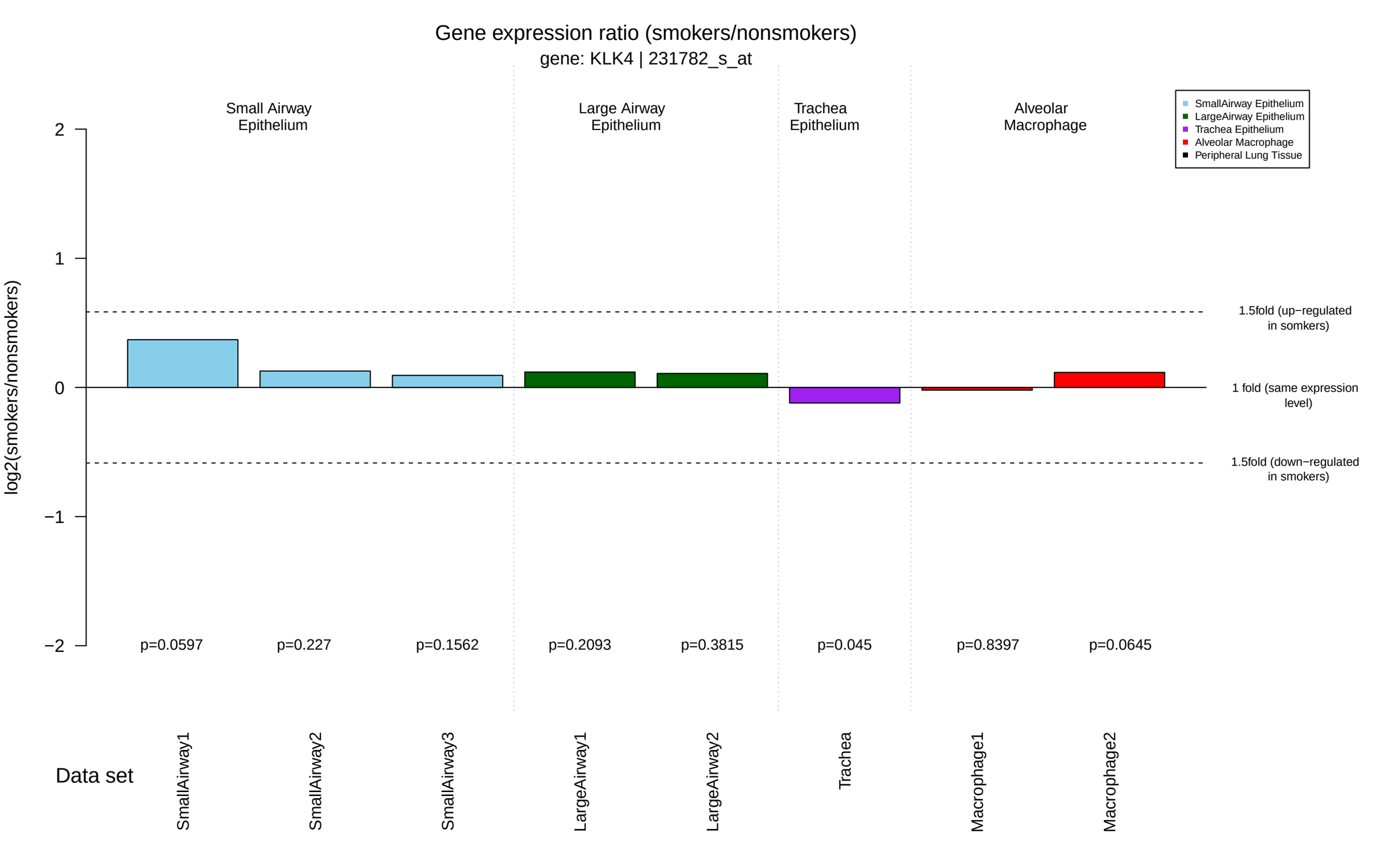
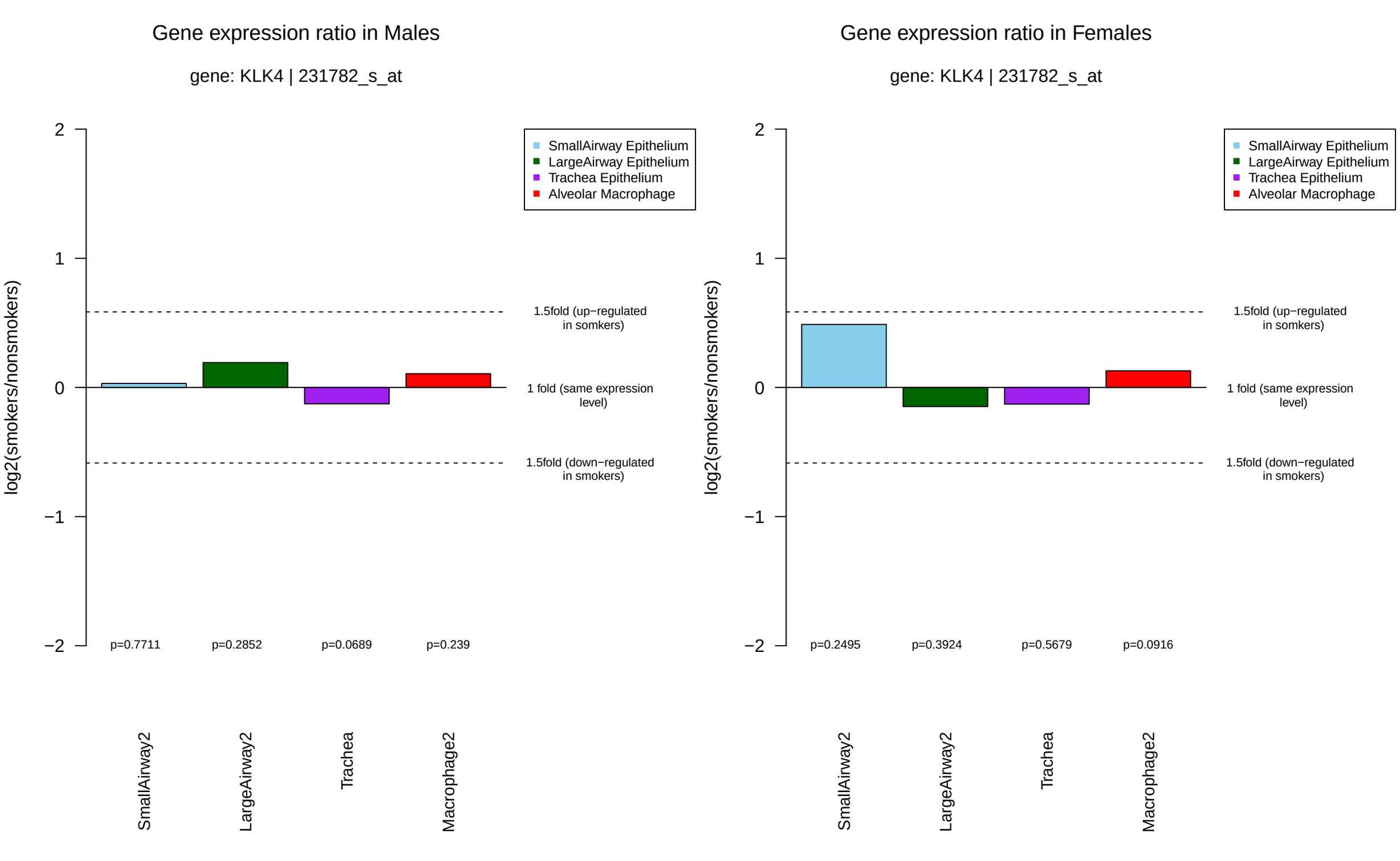
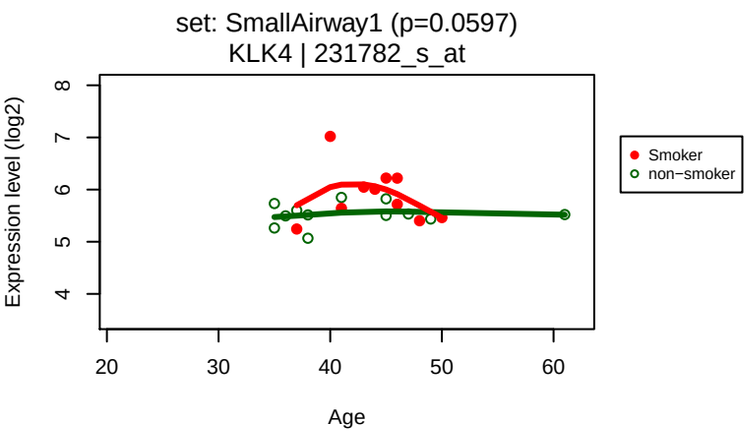
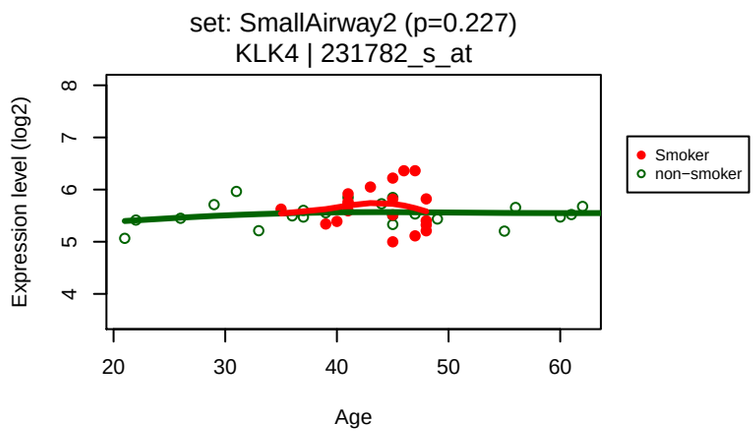
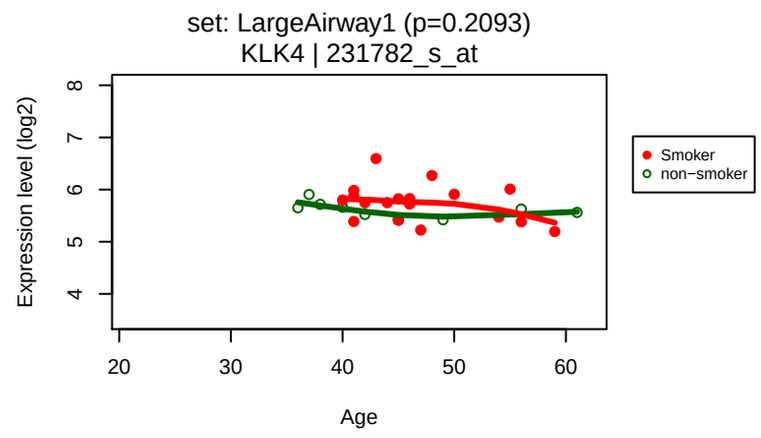

Gene expression of smokers and nonsmokers
gene: KLK4 | 231782_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0597
p=0.227
p=0.1562
p=0.2093
p=0.3815
p=0.045
p=0.8397
p=0.0645
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

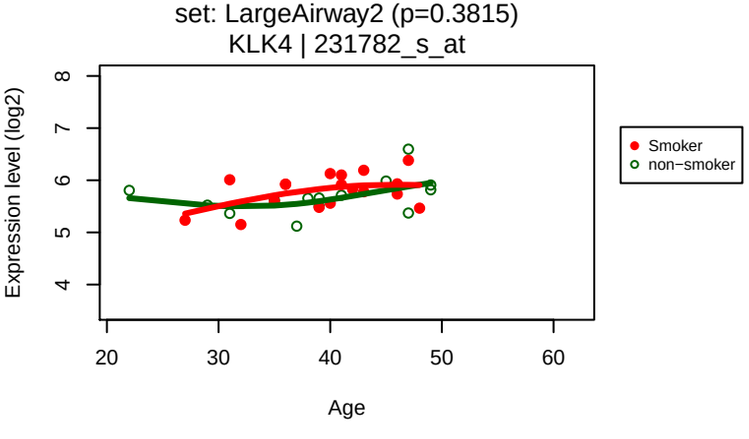

Gene expression ratio (smokers/nonsmokers)
gene: KLK4 | 231782_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0597
p=0.227
p=0.1562
p=0.2093
p=0.3815
p=0.045
p=0.8397
p=0.0645
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

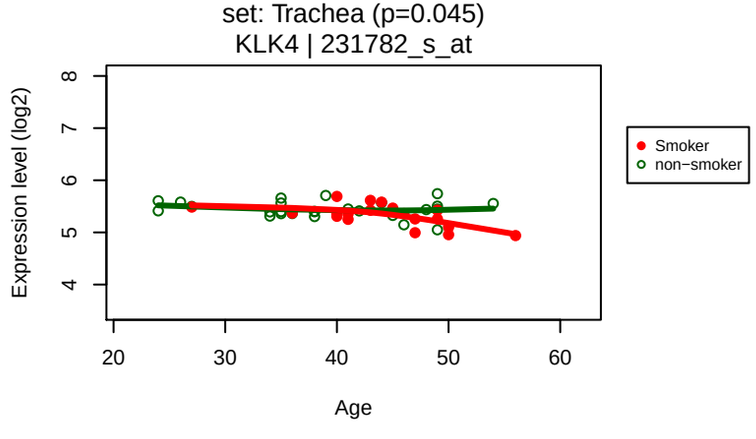

Gene expression ratio in Males
gene: KLK4 | 231782_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7711
p=0.2852
p=0.0689
p=0.239
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLK4 | 231782_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2495
p=0.3924
p=0.5679
p=0.0916
SmallAirway2
LargeAirway2
Trachea
Macrophage2

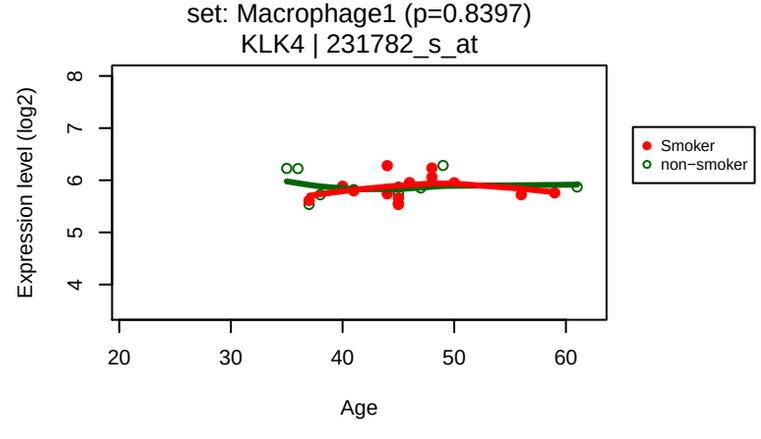

set: SmallAirway1 (p=0.0597)
KLK4 | 231782_s_at
Expression level (log2)
Age
Smoker
non−smoker

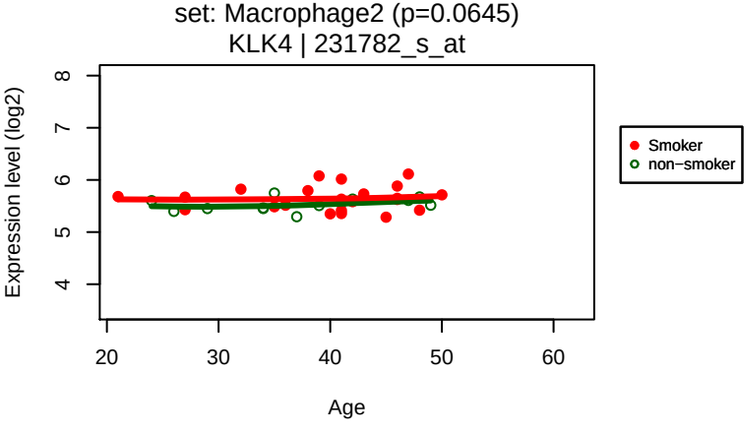

set: SmallAirway2 (p=0.227)
KLK4 | 231782_s_at
Expression level (log2)
Age
Smoker
non−smoker

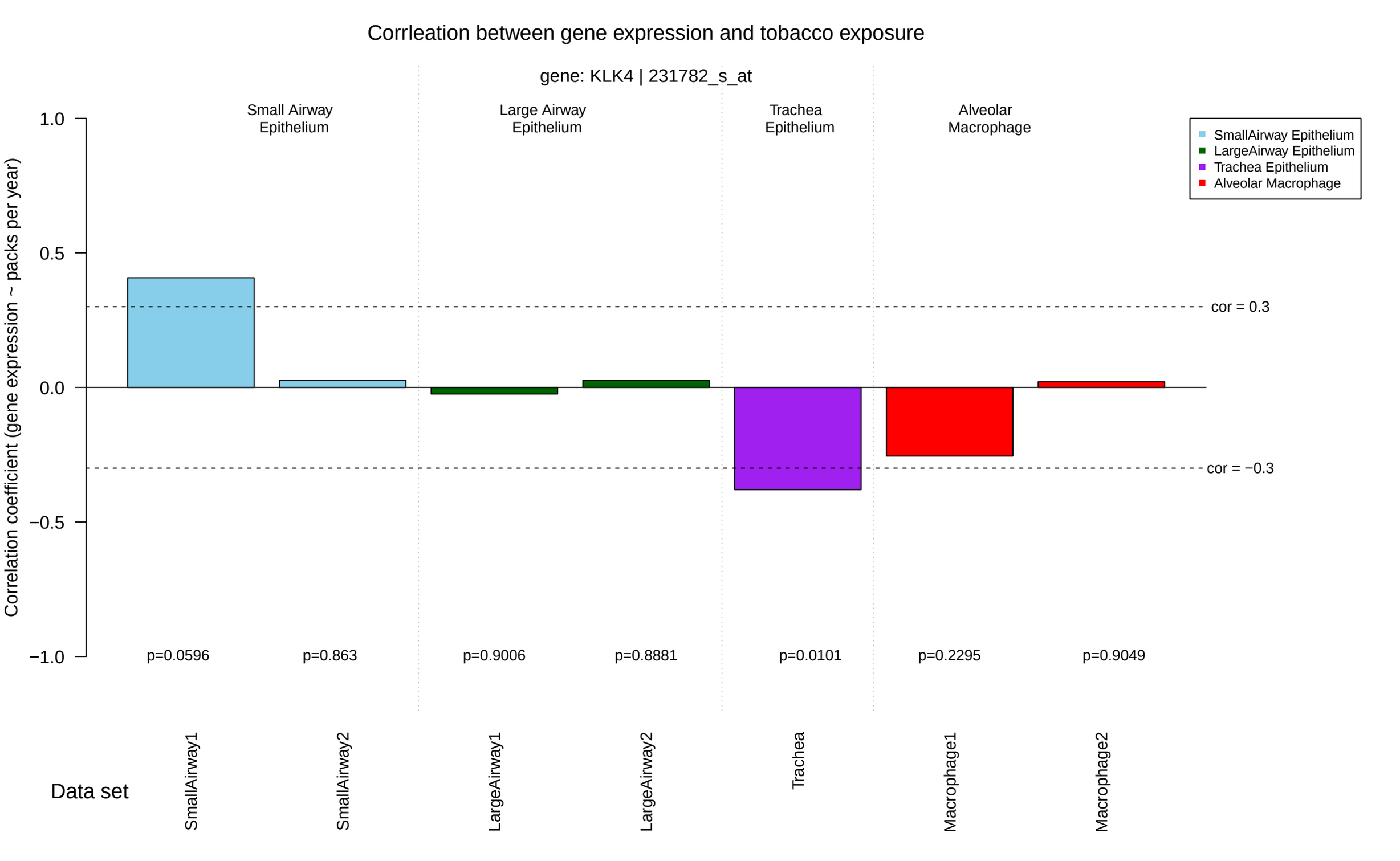

set: LargeAirway1 (p=0.2093)
KLK4 | 231782_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3815)
KLK4 | 231782_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.045)
KLK4 | 231782_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8397)
KLK4 | 231782_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0645)
KLK4 | 231782_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLK4 | 231782_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0596
p=0.863
p=0.9006
p=0.8881
p=0.0101
p=0.2295
p=0.9049
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2